

RUSHI ASHOK SAVANI

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WORK EXPERIENCE

AI Engineer / Data Science Co-op

Jun 2025 – May 2026

ProHealth Home Health and Hospice

Atlanta GA

- Built a 4-agent **LangGraph** chatbot using **Snowflake Cortex**, with buffer memory and a feedback loop for error handling, unifying fragmented patient data and cutting administrator workload by 20+ hours weekly.
- Diagnosed semantic-retrieval failures using **RAGAS** against a 50+ question eval set, fixing them with hybrid retrieval to lift accuracy 35% and answer rate to 90%.
- Resolved class imbalance in the **XGBoost** readmission model using native **class weight** instead of SMOTE to avoid synthetic-data audit risk in a regulated setting, lifting recall from **48% to 76%**.
- Designed three **LangChain**-based data pipelines generating citation-backed root cause reports using **LLM** for flagged cases clinicians could not manually investigate, which surfaced about **\$20K** in preventable opportunities during Q1.
- Selected models by A/B testing response quality with clinicians and scoring outputs against a custom rubric using LLM-as-judge, validating outputs against real clinical judgment before deployment.
- Audited 200+ pipeline runs manually via **DeepEval**, reviewing response quality at each step alongside latency, cost, and token usage to assess the system's operational cost-effectiveness.
- Deployed **SageMaker** endpoints and the investigation pipeline via Dockerized FastAPI on **AWS EC2** with GitHub Actions CI/CD, and CloudWatch monitoring caught a **10% recall drift** in production with zero downtime.

Data Science Research Assistant

Jan 2025 – May 2025

Syracuse University

Syracuse, NY

- Automated a **Python** pipeline that handled missing values, standardized unstructured text and normalized mixed language responses into English before fine-tuning, saving 15+ hours.
- Fine-tuned RoBERTa in **PyTorch** via Hugging Face Transformers on ~7,000 survey responses, tuning optimal learning-rate through loss curve with AdamW, reaching **91%** accuracy under stratified K-fold validation.
- Used Sentence-BERT embeddings and HDBSCAN to cluster AI adoption themes, showing visualization in **Streamlit** dashboard built for non-technical research stakeholders.

Data Analyst

Jun 2023 - Jun 2024

Krishna Jewels

Mumbai, IN

- Migrated 40K+ paper-based inventory, sales, and supplier records into a normalized **MySQL** schema, cutting record lookup time by **95%** through indexing and query rewrites.
- Automated 15+ hours of weekly manual analysis by building **SQL ETL pipelines** into **Power BI** with DAX measures for sales, margin, and inventory KPIs, building interactive dashboards.

TECHNICAL SKILLS

- Languages & Data:** Python, SQL, PostgreSQL, MySQL, TypeScript, Git, REST APIs
- LLM & GenAI:** LangChain, LangGraph, MCP, RAG, Vector Databases, Prompt Engineering, DeepEval, RAGAS
- Machine Learning:** Scikit-learn, NumPy, XGBoost, Classification, Regression, MLflow, SHAP, Cross Validation, A/B Testing
- Deep Learning & NLP:** PyTorch, TensorFlow, Hugging Face Transformers, RoBERTa, ClinicalBERT, Sentence-BERT, TF-IDF
- BI & Analytics:** Power BI, Tableau, Streamlit, ETL/ELT, Data Modeling, Statistical Analysis, Hypothesis Testing
- MLOps & Cloud:** AWS SageMaker, Snowflake, Docker, FastAPI, GitHub Actions, CI/CD, CloudWatch, Model Monitor

EDUCATION

Syracuse University, NY, USA

Aug 2024 - May 2026

Masters | Information Systems, Data Science

(GPA: 3.86/4)

- Relevant Coursework:* Applied Machine Learning, Database Management, Data Science, Cloud Management

Thadomal Shahani Engineering College, Mumbai, India

Aug 2020 - Jun 2024

Bachelors | Information Technology

(GPA: 8.52/10)

PROJECTS

PrivateRAG | *Fully Local Private-Document Q&A System*, [GitHub](#)

- Designed a fully local **Retrieval-Augmented Generation (RAG)** stack with document ingestion, Qdrant vector indexing, metadata filtering, and **Llama 3.1** inference with zero external API calls.
- Wrote a **TypeScript** chat interface over a local OpenAI-compatible backend with streaming responses, source attribution, and chunk-level traceability.

AI Email Response System | *Autonomous Emailing Pipeline*, [GitHub](#)

- Cut response wait time 90% by engineering a RAG email-reply generator that checks escalation rules first, drafting a hand-off instead of an autonomous resolution for high-risk cases.
- Reduced human touch on inbound support email by auto-routing policy-clean, high-confidence replies through a provider-agnostic **MCP** client, escalating only cases a ground-truth-validated judge flags as policy-mandated.

COVID-19 India Analytics | *Interactive Tableau Dashboard*, [Dashboard](#)

- Built a **Python** pipeline to reconcile inconsistent state-name spellings across public datasets, powering an interactive **Tableau** dashboard that flagged states with high testing but low vaccination coverage.